Hengyu Jin

EDUCATION

Tongji University (TJU)
Degree: B.Eng. in Software Engineering | **GPA:** 4.60/5.00 (91.01/100) Sep 2023 – Jun 2027 (expected)

Research Interests: Reasoning and Reliability in Foundation Models; AI Agents; Human-AI interaction

PUBLICATIONS & PREPRINTS

- **H. Jin**, S. Yang, and D. Wang. "Final Checkpoints Are Not Enough: Analyzing Latent Reasoning Faithfulness Along Training Trajectories." *Under review at ACL Rolling Review*. [arXiv]
- **H. Jin**, S. Yang, and D. Wang. "Concord: An Interactive Environment for Evaluating Multi-User Agents under Distributed Information and Consent." *Under review at ACL Rolling Review*.
- W. Ye, X. Yuan, Y. Bin, **H. Jin**, L. Peng, P. Zeng, and H. T. Shen. "RISER: Orchestrating Latent Reasoning Skills for Adaptive Activation Steering." *Findings of ACL 2026*. [Paper]
- J. Du, Y. Zhang, **H. Jin**, X. Zou, J. Zhu, and X. Fu. "Kairotask: Probing the Bridge Between Vague Intents and Spatiotemporal Contexts." *CHI EA 2026*. [DOI]
- T. Chen, Z. Li, X. Chen, Y. Shen, **H. Jin**, L. Zhang, and S. Zhao. "CORA: A Cognitive Reframing Dialogue Agent for Supporting Emotion Regulation." *IEEE Transactions on Affective Computing*, 17(3): 2432–2447, 2026. [DOI]
- L. Mao†, J. Cao†, L. He†, W. Xu, Y. Yan, **H. Jin**, and 19 others. "RobotAgentBench: Towards Open-world Robot Autonomy." *arXiv preprint*. [arXiv] (†equal contribution)

RESEARCH EXPERIENCE

PRADA Lab, KAUST Mar 2026 – Oct 2026
Visiting Student | Advisor: **Prof. Di Wang** *Saudi Arabia & Remote*

Mechanistic Analysis of Latent Reasoning Faithfulness

- Led a first-author investigation of how latent reasoning faithfulness evolves throughout training, showing why final-checkpoint performance alone can mask changes in a model's internal reasoning process.
- Built a ProsQA evaluation pipeline combining BFS-verified counterfactual edits with latent-state interventions to test whether latent reasoning states causally reflect changes in problem structure across checkpoints.
- Compared latent and explicit reasoning approaches across diverse tasks, identifying distinct training dynamics that inform more reliable evaluation of reasoning systems.

Concord: Evaluating Multi-User Agents under Distributed Information and Consent

- Led a first-author study of LLM agents that serve several interdependent users, formalizing two forms of task interdependence: distributed information and distributed consent.
- Built Concord, an interactive environment with 309 tasks across lodging, dining, and shopping on real catalogs and 190 groups of simulated users; agents read personal records, message users, obtain consent, and place orders whose executed final state determines success.
- Evaluated thirteen LLMs and analyzed their failures and trajectories, showing that the main bottleneck is obtaining users' consent rather than gathering their information, and that similar success rates can hide very different costs to the users.

Aether AI Jun 2026 – Present
Research Intern | Advisor: **Prof. Biwei Huang (UC San Diego)** *Remote*

Robot-Use Agents System

- Co-authored RobotAgentBench, a benchmark for open-world robot autonomy in realistic household environments, contributing to the construction of its simulation data. [arXiv]
- Developing an execution-time verifier for a robotics agent system to detect failures during VLA execution and trigger early termination, replanning, or tool-assisted recovery.
- Built a natural failure-data generation pipeline in Genie Sim by identifying atomic subtasks from simulator states, checkpointing key execution states, and resuming rollouts with weaker VLA policies to produce realistic failure trajectories.

Tongji University Oct 2024 – Dec 2025
Undergraduate Researcher *Shanghai*

RISER: Hidden State Steering to Enhance LLM Latent Reasoning Skills

- Co-developed a plug-and-play activation-steering framework that dynamically injects cognitive-primitive vectors into intermediate layers of frozen LLMs, achieving average accuracy gains of 3.4–6.5 percentage points over the base models and 2–3× higher token efficiency than chain-of-thought baselines.
- Contributed to building a cognitive primitive construction pipeline based on Contrastive Activation Addition (CAA) + LLM-as-Judge filtering + KMeans clustering, distilling six categories of reusable reasoning vectors from Transformer hidden states.

CORA and Follow-up Work on Emotional Support | Advisor: **Prof. Ying Shen**

- Designed and implemented a Streamlit-based evaluation platform for blinded comparisons across model variants and configurable inference settings, supporting experimental validation and dialogue-data collection.
- In follow-up work on emotional support with different personality styles, used LLaMA-Factory to fine-tune Qwen2.5-7B-Instruct with LoRA on CBT-guided dialogues.

Tsinghua University Jun 2025 – Oct 2025
Research Intern | Advisor: **Prof. Xinyi Fu** *Remote*

Kairotask: Context-Aware Scheduling from Vague Intentions

- Developed a Qwen-Plus-based backend to translate vague scheduling requests into structured temporal metadata.
- Led the development of a complete mobile application by integrating iOS Core Location with geofencing technology, achieving location-based context-triggered notifications and reminders, and supporting an eight-day field deployment study.

SELECTED PROJECTS

Co-founder & Full Stack Development Lead | Follo AI: Context-Aware Scheduling Agent Nov 2025 – Feb 2026

- Extended the Kairotask framework into a context-aware scheduling agent, designing a response–perception–memory architecture that combines LLM-based intent interpretation, environmental context, and structured user memory.
- Led end-to-end development of the mobile application and backend, delivering a functional MVP for proactive task management, context-aware recommendations, and real-world reminder triggering.

Team Lead & Core Developer | "Scrollscape" Vision Pro AR Application Jun 2025 – Sep 2025

- Led the development of an immersive Chinese-garden education application for Apple Vision Pro, translating architectural and aesthetic concepts into interactive spatial-computing experiences.
- Implemented visionOS interaction and rendering components, including gesture-based navigation, virtual–physical scene transitions, and an automated aesthetic-evaluation module.

AWARDS AND HONORS

First Prize (East China Region), China Collegiate Computing Contest–Mobile App Innovation Division (2025)
First-grade Ethnic Class Special Scholarship, TJU (2023–2024, 2024–2025)

SKILLS

Relevant Coursework: Machine Learning, Computer Vision, Discrete Mathematics, Linear Algebra, Probability and Statistics
Languages: Chinese (Native); English (TOEFL iBT: 4.5/6)
Programming: Python, C++, Swift
ML & Research Tools: PyTorch, Hugging Face Transformers, LLaMA-Factory
Development: SwiftUI, React, FastAPI
Systems & Tools: Git, LaTeX, Linux, Slurm